

Figure 1 is a contour plot of the function $f(x, y) = 100 - 0.5x^2 - 0.5y^2$. The plot shows a central peak at $(0, 0)$ with a value of 100. The contours are concentric ellipses centered at the origin. The x and y axes range from -20 to 20. A legend indicates that solid lines represent positive values, dashed lines represent negative values, and a thick solid line represents the zero contour.

